

EBRAHIM ALIYOU WUDU

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RESEARCH INTERESTS

Mechanistic interpretability, representation learning, LLM evaluation and alignment, self-supervised learning, efficient multimodal AI, and robust AI systems. Interested in empirical research that isolates causal mechanisms, tests hypotheses through controlled experiments, and produces reproducible evaluation.

RESEARCH EXPERIENCE

AI Research Intern

May 2026 – Present

University of Galway / Part-Time, Remote

Supervisor: Dr. Saeed Alsamhi

sentinelMAS – Evidence-Guided Multi-Agent AI Framework for Histopathology Classification

- Designed an evidence-guided adaptive cascade combining EfficientNet-B0, TinyViT, and the frozen UNI pathology foundation model, with calibrated confidence, conformal uncertainty, and learned modality gating for selective escalation.
- Achieved **88.77% external accuracy** on the patient-disjoint CRC-VAL-HE-7K cohort, compared with 78.59% for a standalone CNN and 88.22% for brute-force ensembling, while reducing total cohort inference latency by **~35%**.
- Ran component ablations, corruption tests, and multi-method explainability analyses using Grad-CAM++, attention rollout, and SHAP to identify branch contributions and robustness under distribution shift.

Paper: sentinelMAS

AI Research Intern

May 2026 – Present

Haqila Lab, Jimma University / Part-Time

Advisor: Dr. Gelan Ayana Zewdie

Self-Supervised ECG Representation Learning Under Graded Noise

- Benchmarking contrastive, reconstructive, and latent-predictive self-supervised learning methods for ECG representation learning under progressively graded physiological and environmental artifacts.
- Evaluating robustness, calibration, representation stability, and latent-space behavior to study how increasing noise alters learned physiological representations.

INDEPENDENT RESEARCH

Steering the Machine: Latent State Manipulation in Gemma-3-12B

Aug 2026

Digital Minds Research Sprint, Apart Research / Independent Research

- Investigated whether latent affective steering produces controlled semantic changes or broader behavioral degradation by intervening in late-layer residual streams (Layers 32 and 40) of Gemma-3-12B.
- Extracted and injected steering vectors at intervention strengths $\alpha \in \{0.0, 0.05, 0.1\}$ with magnitude-matched random-vector controls; evaluated self-reported valence, refusal behavior, sentiment, and GSM8K reasoning under a 256-token generation constraint.
- Observed semantic shifts under affective steering, while high-magnitude random perturbations caused token looping and structural collapse; refusal-oriented steering frequently produced hyper-rationalization rather than explicit refusal.
- Owned hypothesis design, experiment implementation, representation-engineering code, quantitative analysis, and manuscript preparation.

Preprint: paper **Code:** GitHub

SELECTED AI/ML PROJECTS

Kuza AI – Offline Agricultural AI Assistant

2026

- Co-founded and engineered an English/Swahili agricultural AI system for East African farmers, focusing on low-resource inference and deployment under constrained connectivity, memory, and compute.
- Built multilingual supervised fine-tuning, data-cleaning, model-adaptation, and local-inference pipelines; benchmarked practical deployment trade-offs for edge-oriented AI.

Project: Devpost **Models/Data:** Hugging Face **Code:** GitHub

EDUCATION

Hindustan Institute of Technology and Science

2021 – 2025

Bachelor of Technology in Computer Science and Engineering, Chennai, India

CGPA: 8.79/10 | First Class with Distinction

Relevant coursework: Machine Learning, Deep Learning, Algorithm Design, Data Warehousing and Mining, Big Data Analytics, Theory of Computation, Computer Organization and Architecture, Cloud Deployment

ENGINEERING EXPERIENCE

Co-Founder & Lead Engineer – Gobez

May 2026 – Present

Addis Ababa, Ethiopia

gobez.cloud

- Led engineering for an EdTech platform serving **500+ active monthly students across 7 schools**; built recommendation and Topic Mastery analytics systems.
- Developed RAG-based semantic search over **5+ years of solved national examination questions**.

Software Development Engineer – Reelo

Apr 2025 – Jan 2026

Remote, India

- Built AWS-backed backend infrastructure processing **100,000+ data points/month** and integrated B2B subscription workflows across Chargebee, Stripe, and Razorpay.

Data Engineer Intern – Renan Partners

Feb 2025 – Mar 2025

Remote, India

- Built multi-client ELT pipelines using Meltano, Airbyte, AWS EC2, Shopify extraction, and dbt transformation workflows.

Backend Developer Intern – Reelo

Dec 2023 – May 2024

Remote, India

- Refactored Node.js APIs, reducing response time by **25%**; implemented authentication and role-based access control.

AWARDS & HONOURS

Study In India Scholarship, Government of India – 2021: Full merit-based government scholarship covering tuition and living costs.

Hackathon for Climate Action – WTUN – 2nd Place, 2023: Waste2Wealth circular-economy prototype.

Hack-o-hits Hackathon – 3rd Place, 2023: Wearable-assisted medicine intake tracking system.

TECHNICAL SKILLS

Research/ML: PyTorch, TensorFlow, scikit-learn, self-supervised learning, representation learning, interpretability, uncertainty estimation, model evaluation, PEFT/QLoRA, RAG

LLM Research: Representation engineering, steering vectors, LLM evaluation, LangChain, LangSmith

Programming: Python, C++, JavaScript, SQL, GraphQL

Systems: FastAPI, Node.js, AWS, PostgreSQL, MongoDB